

Robust Treatment of Degenerate Elements in Interactive Corotational FEM Simulations

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Abstract

We address the problem of robust and efficient treatment of element collapse and inversion in corotational FEM simulations of deformable objects in two and three dimensions, and show that existing degeneration treatment methods have previously unreported flaws that seriously threaten robustness and physical plausibility in interactive applications. We propose a new method that avoids such flaws, yields faster and smoother degeneration recovery and extends the range of well-behaved degenerate configurations without adding significant complexity or computational cost to standard explicit and quasi-implicit solvers.

Keywords: deformable objects, finite element method, interactive simulation

ACM CCS: Categories and Subject Descriptors (according to ACM CCS): I.6.8 [Simulation and Modelling]: Types of Simulation Animation; I.3.5 [Computer Graphics]: Computational Geometry and Object Modelling Physically based modelling; I.3.7 [Computer Graphics]: Three-Dimensional Graphics and Realism Animation and Virtual Reality.

1. Introduction

Simulation of elastically deformable objects is an important problem in many areas such as structural analysis, computer-generated feature films, virtual reality and videogames. Among possible computational schemes, the finite element method (FEM) is the most physically accurate, as it is based on continuum mechanics and observable material parameters.

In the context of interactive simulation, physical realism is of foremost importance, but limited by strict computational efficiency and numerical stability requirements. Over the last decade specific simulation schemes using the corotational linear FEM (CLFEM) formulation have been developed that prioritize efficiency and robustness over realism or accuracy while yielding visually plausible results.

Interactive simulation methods must guarantee robustness under unexpected circumstances, as opposed to offline methods, in which the simulation context is known *a priori* (structural analysis) or

can be adjusted and post-processed to eliminate displeasing results [conjugate gradient [CG] film production). Robustness is critical for commercial videogames, where the possibility of crashes or visual errors is enough to discard a feature. In CLFEM, one source of such problems is element degeneration (collapse and inversion), unavoidable due to unpredictable events like offending user interactions or violent collisions. Several methods have been proposed to deal with element degeneration.

In this work, we begin with a description of the CLFEM formulation (Section 3) and the robustness and plausibility issues resulting from commonly used degeneration treatment methods (Sections 4.1 and 4.3). In Section 5, we propose a new method that computes a continuous-time degeneration direction that eliminates the potential inaccuracy of discrete-time heuristic approaches (Section 5.1) and performs an inexpensive rotation correction using such direction to improve degeneration response (Section 5.2). We compare our method with the existing alternatives in Section 6, and report the final conclusions in Section 7.

2. Related Work

In computer graphics, deformable solid simulation based on non-linear continuum mechanics was pioneered by [TPBF87], using a finite difference material discretization with semi-implicit time integration. A linear FEM (LFEM) scheme with explicit integration was presented in [OH99] to simulate plausible solid fracturing, limited to small deformations. An adaptative FEM simulation method was introduced in [DDCB01], using explicit integration to achieve real-time haptic interaction.

Stable interactive simulation of free deformable solids became possible when stiffness warping was introduced in [MDM*02] and later improved by the now popular corotational method in [MG04], [HS04]. Implicit integration with large timesteps [BW98] was favoured to prioritize efficiency and stability over accuracy. Element inversion was soon identified as a potential threat to robustness and plausibility, and treated using an anisotropy-inducing QR decomposition in [NPF05]. A similar parallelized algorithm was presented in [PO09] and used in commercial videogames. The limitations of quasi-implicit integration have been recently overcome by fully implicit non-linear corotational methods built on the evaluation of exact force differentials in [CPSS10], [MZS*11] and [Bar12].

An isotropic scheme for invertible finite element simulation of hyperelastic materials was introduced in [ITF04], using the singular value decomposition (SVD) and explicit integration of the elastic forces. This scheme was successively extended in [TSIF05], [GKJD09] and [SZL*11] to support fully implicit integration, and culminated in an energetically consistent method in [SHST12]. A method for isotropic invertible CLFEM was presented in [ST08], building on [ITF04] and enforcing time-consistent uninversion directions. Recently, a comparison between invertible FEM schemes for computer graphics has been published [SSB13].

3. Interactive FEM Simulation

One of the most advanced methods for interactive simulation of deformable solids is CLFEM. It combines the physical realism, simplicity and efficiency of LFEM with the support for large deformations provided by shape-matching simulation schemes. Nearly unconditional stability can be achieved by means of quasi-implicit integration, and standard iterative solvers offer an accuracy/speed tradeoff suited to realtime simulation, as well as opportunities for parallelization on modern hardware architectures.

3.1. Linear FEM (LFEM)

LFEM is the simplest possible FEM scheme, as it uses simplex elements (triangles in two dimensions, tetrahedrons in three dimensions) with linear shape functions and a linear constitutive model [ESHD05]. This basic formulation yields a linear relationship between element node displacements and the resulting elastic forces $f_e = -K_e(x_e - r_e)$, where $r_e = [r_1^T \dots r_4^T]^T$ and $x_e = [x_1^T \dots x_4^T]^T$ are the reference and deformed element node positions, respectively, $f_e = [f_1^T \dots f_4^T]^T$ are the elastic forces on the element nodes and K_e is the constant stiffness matrix. For a finite element mesh,

the equations of motion of all nodes can be assembled:

$$\begin{aligned} \mathcal{M}\ddot{x} &= f, \\ f &= f_{\text{ext}} - \mathcal{C}\dot{x} - \mathcal{K}(x - r), \end{aligned}$$

where $x, \dot{x}, \ddot{x}$ are the deformed node positions, velocities and accelerations, r are the reference node positions, f_{ext} are the external forces that act on the nodes, $\mathcal{M}$ is the system mass matrix, usually diagonally lumped for efficiency, $\mathcal{C}$ is the damping matrix, usually derived from Rayleigh damping $\mathcal{C} = \alpha\mathcal{M} + \beta\mathcal{K}$ and $\mathcal{K}$ is the assembled system stiffness matrix that aggregates the relationship between all finite element elastic forces and all node displacements [ESHD05].

Dynamics can be integrated implicitly to achieve nearly unconditional stability using Backwards–Euler [BW98]:

$$\left(\mathcal{M} - \Delta t \frac{\partial f}{\partial \dot{x}} - \Delta t^2 \frac{\partial f}{\partial x}\right) \Delta v = \Delta t \left(f + \Delta t \frac{\partial f}{\partial x} \dot{x}\right).$$

The required elastic and damping force differentials $\frac{\partial(-\mathcal{K}(x-r))}{\partial x} = -\mathcal{K}$ and $\frac{\partial(-\mathcal{C}\dot{x})}{\partial \dot{x}} = -\mathcal{C}$ are constant and therefore may be pre-computed, which reduces the computational cost at runtime. Moreover, all involved matrices are sparse and efficient iterative linear system solvers such as the CG can be used, requiring only sparse matrix–vector products and sums. Implicit integration allows large timesteps, up to 1/60s or even 1/30s, at the cost of increased numerical damping, which is an acceptable drawback in most interactive entertainment applications [PO09].

3.2. Corotational LFEM

Due to the linearity of the deformation measure, LFEM cannot realistically model solids that undergo significant shape changes and rotation. The key idea behind CLFEM is the extension of LFEM to achieve elementwise rotational invariance, evaluating the node displacements in a rotated coordinate frame $\mathcal{R}$ that matches the deformed element configuration and the reference one [MG04, HS04].

In CLFEM, LFEM forces are computed locally to the element and rotated to the global coordinate frame according to $f_e = -\mathcal{R}_e K_e (\mathcal{R}_e^T x_e - r_e)$, where $\mathcal{R}_e$ is an element rotation matrix formed by repeated $\mathcal{R}$ diagonal blocks. Alternatively, we can consider that the reference configuration was rotated to match the deformed one, so that $f_e = -K_e^R (x_e - \mathcal{R}_e r_e)$ with $K_e^R = \mathcal{R}_e K_e \mathcal{R}_e^T$.

Even with a linear material model, corotational methods are geometrically non-linear, and their fully implicit integration requires the evaluation of exact force differentials as well as a non-linear equation solver. For interactive simulation, a common simplification uses quasi-implicit integration with inexact force differentials $\frac{\partial f_e}{\partial x_e} \approx -K_e^R$ that drop all terms that depend on $\frac{\partial \mathcal{R}}{\partial x_e}$, effectively considering $\mathcal{R}$ constant during each timestep. This approach is widely used in interactive applications, as it only requires a single linear equation solver (CG) invocation per timestep, but is known to cause noticeable energy loss [CPSS10] and potential artefacts in case of large stretch deformation [MZS*11, SHST12].

Following the finite volume notation introduced in [TBHF03] we define the piecewise linear deformation mapping from the undeformed element configuration to the deformed one as $\phi(r) = \mathcal{F}r + b$. Matrix $\mathcal{F}$ represents the translation-independent element deformation matrix $\mathcal{F} = \mathcal{D}_s \mathcal{D}_m^{-1}$, where $\mathcal{D}_m = [r_{12} \ r_{13} \ r_{14}]$ and $\mathcal{D}_s = [x_{12} \ x_{13} \ x_{14}]$, with reference and deformed edge vectors $r_{ij} = r_j - r_i$ and $x_{ij} = x_j - x_i$, respectively. The global translation vector b does not affect the element shape. A best matching rotation $\mathcal{R}$ can be extracted from the deformation $\mathcal{F}$. This is a general problem that appears in different areas (CLFEM, meshless shape-matching, computer vision). Several numerical methods to extract $\mathcal{R}$ from $\mathcal{F}$ have been developed and used in CLFEM: QR factorization [NPF05],[PO09], polar decomposition [MG04],[PO09] and SVD [ITF04]. We will detail them in Section 4.

3.3. Element degeneration

Element collapse and inversion is practically unavoidable in interactive simulation and threatens numerical robustness and physical realism. Elements may temporally or persistently collapse and invert due to strong external forces or contact constraints. Degeneration may happen due to geometry discretization even if the element nodes follow their ‘correct’ continuum trajectories [ITF04].

For tetrahedral elements in three dimensions, degeneration can be easily detected by computing the volume of the deformed element $V_s = \frac{1}{6} \det(\mathcal{D}_s)$ (and similarly for triangle elements in two dimensions). The element is *collapsed* if $V_s \approx 0$, and *inverted* if $V_s < 0$. Assuming a non-degenerate reference configuration with volume $V_m = \frac{1}{6} \det(\mathcal{D}_m) > 0$, it can be shown that $\det(\mathcal{F})$ is actually the volume change fraction from reference to deformed configurations, as $\det(\mathcal{F}) = \det(\mathcal{D}_s \mathcal{D}_m^{-1}) = \det(\mathcal{D}_s) / \det(\mathcal{D}_m) = V_s / V_m$. Noting that $\text{sign}(\det(\mathcal{F})) = \text{sign}(\det(\mathcal{D}_s))$. This is particularly useful to detect collapse and inversion in a scale-independent way. Geometrically, tetrahedral elements collapse when any pair of features (Vertices, Edges, Faces) become coincident, yielding a singular $\mathcal{F}$. The basic collapse cases are: V–F, V–E, V–V and E–E. For triangles in 2D, only V–E and V–V cases are possible. Numerically, *collapsed* elements result in a singular $\mathcal{F}$ with $|\det(\mathcal{F})| \leq \epsilon$ and *inverted* elements result in a non-singular $\det(\mathcal{F}) < -\epsilon$ for some tolerance $\epsilon > 0$.

Preventing degeneration in all dynamic situations would require unbounded elastic forces, with the associated numerical integration difficulties. Some non-linear material models used in offline simulation [Ogd97] include terms that grow to infinity as the element collapses and require advanced methods with error control and adaptative timestep to be robustly integrated. In practice, this is not acceptable for interactive simulations. Moreover, degeneration would still be possible in the common cases of overconstrained contact with kinematic geometry, scripted or direct user control of specific node trajectories, as well as initially inverted configurations.

4. Degenerate Element Treatment in CLFEM

Successful treatment of degenerate elements must ensure:

- Numerical robustness: Forces must be bounded and continuous to ensure that numerical methods converge.

- Physical plausibility: Forces must not yield unnatural-looking dynamics, jitter or perceivable visual glitches.
- Self-correction: Elements must rapidly recover when the degeneration cause disappears.

LFEM fulfills all these requirements, and it has been shown [ST08] that CLFEM does too in an instantaneously fixed local frame $\mathcal{R}$. In order to extend this result to a time-varying $\mathcal{R}(\mathcal{F}(t))$ we need to ensure that $\mathcal{R}$ can be robustly computed in any configuration and that it yields consistently plausible and self-correcting forces as the element deforms. Throughout this section we analyse in detail the existing methods to extract $\mathcal{R}$ from a potentially degenerate $\mathcal{F}$, and show that none of them fulfills all requirements.

4.1. QR

Given a non-singular $\mathcal{F}$, its QR factorization $\mathcal{F} = \mathcal{R}\mathcal{E}$, where $\mathcal{R}$ is a pure rotation and $\mathcal{E}$ is upper-diagonal, can be computed using Gram–Schmidt orthonormalization. Usually the direction of the first column of $\mathcal{F}$ is chosen and normalized as the first local axis $\hat{a}_1$ of $\mathcal{R}$, the second column of $\mathcal{F}$ is projected and normalized to make it orthonormal to the first one and obtain $\hat{a}_2$ and the third orthonormal direction is obtained by means of a properly oriented cross-product $\hat{a}_3 = \hat{a}_1 \times \hat{a}_2$ [NPF05].

When $\mathcal{F}$ is singular due to element collapse, as long as two columns remain uncollapsed and linearly independent, all axis of $\mathcal{R}$ can be computed using cross products. If only one column is uncollapsed or linearly independent, its corresponding $\hat{a}_i$ is available, but the other two are undetermined and any pair of orthonormal vectors spanning the plane perpendicular to $\hat{a}_i$ can be arbitrarily selected. If all columns of $\mathcal{F}$ are collapsed, $\mathcal{R} = \mathcal{I}$ is often arbitrarily selected.

Overall, QR factorization is robust to degenerate elements and has a small and fixed computational cost for any $\mathcal{F}$, always resulting in a valid rotation $\mathcal{R}$ with $\det(\mathcal{R}) = 1$. However, when combined with local LFEM, QR factorization has numerous drawbacks:

- Anisotropy due to axis ordering causes a bias in force directions that is strongly dependent on the reference configuration and grows unacceptable for large poisson ratios, yielding physically implausible dynamic behaviour. In the right panel of Figure 2 the streamlines show asymmetric trajectories for a symmetric element that induce inversion even for undegenerate configurations.
- Discontinuity across V–E and V–V collapse cases due to the change of column-axis usage order or arbitrary axis selection can cause jitter.
- A critical point at a specific collapse configuration with $\det(\mathcal{F}) = 0$ that dramatically distorts the rotation field (Figure 2) can be proved to exist when the first column f_1 of $\mathcal{F}$ approaches 0. Near the critical point, any slightly inverted configuration results in a self-induced deep inversion and a counterintuitive uninversion trajectory.
- For V–F degenerations in three dimensions and V–E in two dimensions, the node V can reach an equilibrium inverted configuration if all nodes in its opposite face/edge are kinematically constrained or their local elastic forces are balanced by other elements or external forces. In this case, the element will remain inverted.

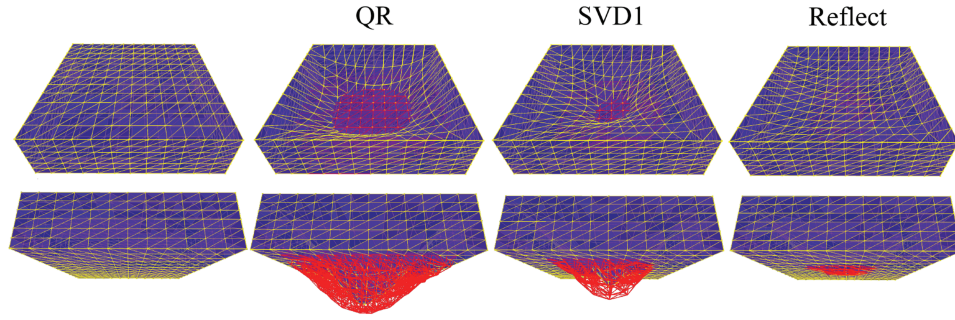


Figure 1: A box supported by all boundary faces except the top ones collapses under gravity. Degenerate elements in red.

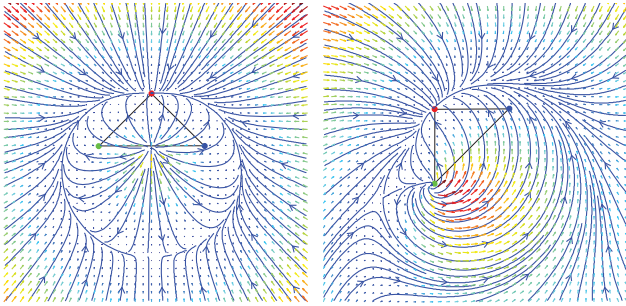


Figure 2: CLFEM force fields experienced by node x_1 (red) in a $[-3, 3]^2$ region of the plane when x_2 (green) and x_3 (blue) are fixed. Field arrows represent force direction and magnitude. Left and right panels show the results of QR for the same element with two different reference orientations. Notice the critical points at the origin (left) and near the green node (right). Material parameters: $E = 1000$, $\nu = 0.35$.

We conclude that QR satisfies the numerical robustness requirements of interactive simulation, but fails to guarantee physical plausibility and self-correction in all situations. The erratic behaviour of QR for a single finite element was mentioned in [NPF05], but its consequences greatly minimized. In Section 6, we will show that QR can introduce macroscopic artefacts on a whole deformable object.

4.2. Polar decomposition

Polar decomposition computes a factorization $\mathcal{F} = \mathcal{R}\mathcal{S}$, where $\mathcal{R}$ is a rotation that minimizes $\|\mathcal{F} - \mathcal{R}\|_F^2$ and $\mathcal{S}$ is a symmetric stretch matrix [SD92]. This factorization is order-independent and does not introduce any anisotropy. In two dimensions $\mathcal{R}$ is directly obtained as:

$$\mathcal{R} = \mathcal{F} + \text{sign}(\det(\mathcal{F})) \begin{bmatrix} \mathcal{F}_{22} & -\mathcal{F}_{21} \\ -\mathcal{F}_{12} & \mathcal{F}_{11} \end{bmatrix}.$$

In three dimensions it can be computed iteratively with $\mathcal{R}_0 = \mathcal{F}$ and $\mathcal{R}_{k+1} = \frac{1}{2}(\mathcal{R}_k + \mathcal{R}_k^{-T})$ until $\mathcal{R}_k \approx \mathcal{R}_{k+1}$, with quadratic convergence for nearly orthogonal $\mathcal{R}_k$ [HRS90], and yields good results

in a few iterations [HS04]. This algorithm works for non-singular $\mathcal{F}$ and its efficiency and isotropy preservation make it an excellent choice for CLFEM when element degeneration is not considered.

However, the method fails for singular $\mathcal{F}$ and returns a reflected rotation with $\det(\mathcal{R}) = -1$ for inverted elements. Reflected rotations effectively reflect the local LFEM forces, encouraging inverted elements to remain inverted [ST08]. In order to ensure numerical robustness and self-correction in all configurations, [PO09] uses QR as an auxiliary method when the volume change fraction is below 6% ($\det(\mathcal{F}) < 0.06$). This is clearly discontinuous when the method used to compute $\mathcal{R}$ changes, and repeats all previously reported drawbacks of QR in inverted configurations.

As a conclusion, in its simple and fastest form, polar decomposition cannot guarantee the robustness, plausibility and self-correction requirements for interactive simulation.

4.3. Singular value decomposition (SVD)

The deformation matrix $\mathcal{F}$ can be decomposed into two rotations and a diagonal stretch matrix using the SVD $\mathcal{F} = \mathcal{U}\hat{\mathcal{F}}\mathcal{V}^T$ [GVL96]

For undegenerate configurations, $\mathcal{U}$ and $\mathcal{V}$ are pure orthonormal rotation matrices with determinant 1 that represent global and material rotations, respectively, and $\hat{\mathcal{F}}$ is a diagonal matrix that contains singular values which represent the principal stretches $\sigma_i > 0$, in non-increasing order $\sigma_1 \geq \dots \geq \sigma_N$ for N spatial dimensions. In this case, it can be shown that $\mathcal{R} = \mathcal{U}\mathcal{V}^T$ is exactly the same rotation computed by the polar decomposition $\mathcal{F} = \mathcal{R}\mathcal{S}$ [HRS90].

However, for inverted configurations with $\det(\mathcal{F}) < 0$ the convention $\sigma_i > 0$ results in a decomposition in which either $\mathcal{U}$ or $\mathcal{V}$ contain a reflection with negative determinant. In [ITF04], a heuristic method is presented to remove the reflection from $\mathcal{U}$, $\mathcal{V}$ by transferring it to the $\hat{\mathcal{F}}$ factor, negating the smallest principal stretch σ_N and its corresponding column vectors in $\mathcal{U}$, $\mathcal{V}$. The rationale behind this heuristic is that the smallest principal stretch determines the minimal inversion direction, and guarantees continuous σ_i across the degenerate region of the element configuration space, as the only modified σ_i is the smallest one σ_N which will evolve from positive (when undegenerate) to zero (when collapsed) and to negative (when inverted), and symmetrically in the reverse case. The collapsed case $\det(\mathcal{F}) = 0$ is handled specifically depending on how

many singular values are zero. Singular vectors associated with zero singular values are undefined but, similar to the QR case, can be constrained to be orthonormal to the non-zero ones. If more than one singular value is zero in three dimensions, two or three arbitrary axes are chosen.

The method presented in [ITF04] uses the corrected $\bar{\sigma}_N, \bar{\mathcal{U}}, \bar{\mathcal{V}}$ in the force computations for any material using the Piola–Kirchhoff formulation, which is enough for explicit integration, while [TSIF05], [GKJD09] and [SZL*11] extended the method to compute the force differentials required for implicit integration. This approach was fully generalized in [SHST12] for any energy-based isotropic material model, including Neo–Hookean materials undefined for negative σ_i , by means of C^1 or C^2 energy extrapolations that allow consistent evaluation of force differentials.

This correction can be also used in CLFEM, computing the reflection-free element rotation matrix as $\bar{R} = \bar{\mathcal{U}}\bar{\mathcal{V}}^T$. This approach, named SVD1, yields the same elastic forces as the Piola–Kirchhoff formulation in [ITF04] with a linear elastic isotropic material model. The rotation $\bar{R} = \bar{\mathcal{U}}\bar{\mathcal{V}}^T$ can be seen as a corrected polar decomposition $\mathcal{F} = \bar{R}\bar{\mathcal{S}}$, where the reflection is embedded in $\bar{\mathcal{S}}$.

This inversion handling heuristic is criticized in [ST08] for being both geometrically inaccurate and time-inconsistent. Geometrically, the smallest σ_i represents the smallest stretch ratio from the reference configuration to the deformed one, which is not necessarily the one associated with the direction that minimizes the distance that a node needs to be moved to uninvert the element, specially for badly shaped elements. Time-inconsistency arises when the shape of the inverted element changes due to internal forces but the element is not yet uninverted, which may yield a smallest σ_N associated with an inversion direction completely different from the shortest one at the previous timestep. They propose a modified method (SVD2) that fixes the geometric inaccuracy by selecting the pair (c_i, v_j) that minimizes the distance that node c_i needs to travel along singular vector v_j to uninvert, and therefore negate the associated σ_j . Time-consistency is enforced by storing the initially selected inversion node c_0 and reusing it to choose instantaneous shortest inversion direction v_j in successive timesteps until the element recovers.

While the problems detected in [ST08] are sound, the proposed corrections do not solve them completely, and may generate force discontinuities that manifest in sudden node accelerations and jitter due to the following issues:

- Initial pair (c_0, v_0) selection uses geometric distance instead of σ_i magnitude. At successive timesteps t_0 and t_1 the same c_0 is reused. If c_0 moves in such a way that different $v_j(t_0)$ and $v_k(t_1)$ are selected, their associated $\sigma_j(t_0)$ and $\sigma_k(t_1)$ will both experience a sudden sign change, which clearly yields discontinuous principal stretches unless both have approximately the same value. This can be seen in the semi-circular discontinuity in the right panel of Figure 3.
- If internal or external interactions (neighbour elements, user input, collisions, etc.) force an inverted element to uninvert during the interval $[t_0, t_1]$ in such a way that c_0 does not cross its opposed face F_{c_0} as expected, the actual uninversion direction will not be the instantaneously selected $v_j(t_0)$, and therefore the associated $\sigma_i(t_0)$ that was negated will become suddenly positive without

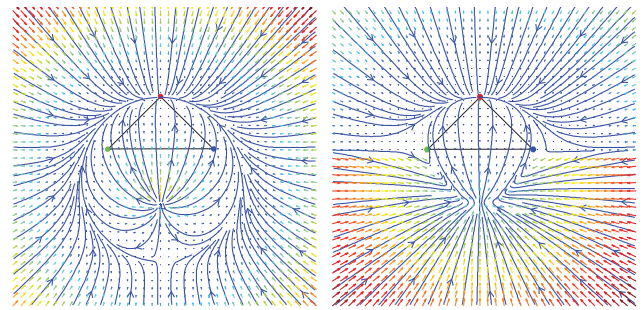


Figure 3: CLFEM force fields experienced by node x_1 (red) in a $[-3, 3]^2$ region of the plane when x_2 (green) and x_3 (blue) are fixed. The left panel shows SVD1 force field (based on [ITF04]), right panel shows SVD2 [ST08]. Node x_1 is selected as c_0 for SVD2. Notice the critical point at $(0, -1)$ in SVD1 (left) and the two discontinuous regions in SVD2 (right), only connected at the critical point with singular values $(1, 1)$. Material parameters: $E = 1000$, $\nu = 0.35$.

passing through 0, which results in a discontinuity in principal stretches. This can be seen in the straight discontinuity across the inversion direction in the right panel of Figure 3.

- Even if time-consistency is enforced afterwards, the initial inversion direction selection heuristic may be wrong for large element shape changes in a single timestep, that is, it may select a direction that does not represent the (vertex, opposite face) pair that actually crossed when considering the continuous node trajectories during the timestep in which the element became inverted (Figure 6).

The method in [ST08] eventually corrects degenerate elements, but the discontinuous σ_i cause visible jitter that compromises physical plausibility. As for corotational elasticity, discontinuities in the principal stretches due to sudden sign flips also produce discontinuities in the associated singular vectors and their derived rotation, which results in discontinuous elastic forces that cause visible jitter.

4.4. Critical points of the SVD

The SVD is widely used, and its analytical and numerical aspects have been thoroughly studied [GVL96], [O’N05], [MS09]. For our purposes, the most relevant result (in [MS09]) states that the reflection-free rotation $\bar{R} = \bar{\mathcal{U}}\bar{\mathcal{V}}^T$ obtained from the SVD of a matrix $\mathcal{F}$ is unique except for two cases:

1. $\text{rank}(\mathcal{F}) < N - 1$;
2. $\det(\mathcal{F}) < 0$ and σ_N is not unique.

The first case arises when $\det(\mathcal{F}) = 0$ and the element is collapsed in more than one direction, which can happen for 2D elements collapsed to a point or 3D elements collapsed to a point or on a line. Fortunately, in the most common case of a flattened element, a single σ_i becomes 0, so $\bar{R}$ is unique and the collapse direction can be safely computed.

The second case is more problematic. For exactly coincident σ_N and σ_{N-1} , the rotation $\bar{R}$ is not unique, but the worst problem

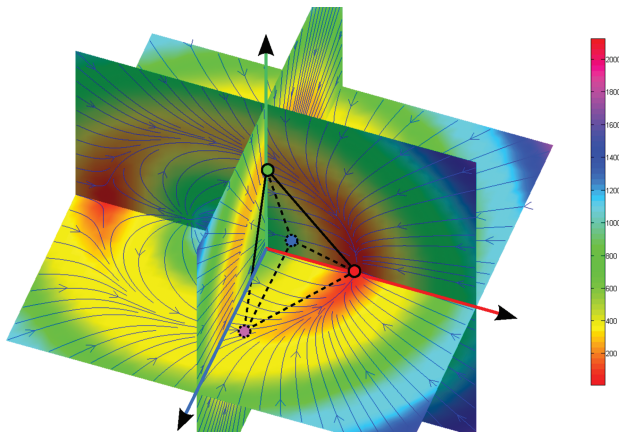


Figure 4: A tridimensional example of the SVD-induced CLFEM force field experienced by node x_1 (red) in a $[-2.5, 2.5]^3$ region of the space when nodes x_2 (green), x_3 (blue), x_4 (violet) are fixed. Element edges shown in black. Field colours represent force magnitude. Arrows represent X (red), Y (green) and Z (blue) axis. Node reference positions are $r_1 = (1, 0, 0)$, $r_2 = (0, 1, 0)$, $r_3 = (0, -\sin 30, -\cos 30)$, $r_4 = (0, -\sin 30, \cos 30)$. Notice the critical point at $(-1, 0, 0)$, similar to the 2D case (left panel of Figure 3). Material parameters: $E = 1000$, $\nu = 0.35$.

is that for different but arbitrarily close singular values a small perturbation to $\mathcal{F}$ can alter singular vectors significantly. These are critical points of the SVD [O’N05]. Numerically, this makes the SVD extremely sensitive to round-off errors around repeated singular values [MS09]. An explicit example is shown in [SD92].

As a result, the singular values σ_i are continuous for perturbations of any $\mathcal{F}$, but the singular vectors are not. When applied to a finite-element deformation matrix $\mathcal{F}$ that evolves in time due to the system dynamics, the second case arises at inverted element configurations where $\det(\mathcal{F}) < 0$, and its practical effects are discontinuous force directions that may compromise implicit integration.

The specific effects on CLFEM are that the reflection-free rotation field $\bar{R}(\mathcal{F}) = \bar{U}\bar{V}^T$ changes rapidly near the critical points of the $SVD(\mathcal{F})$ and covers all possible orientations both in two dimensions (Figure 3) and three dimensions (Figure 4), intuitively wrapping LFEM forces around them in such a way that inversion-increasing forces are generated. In the worst case, equilibrium inverted configurations arise for V–F degenerations in three dimensions and V–E in two dimensions if all nodes except V are kinematic or their local elastic forces are balanced by other elements or external forces. This is also true for non-linear material models, as shown in Figure 5.

Several authors ([GKJD09], [SZL*11], [SHST12]) describe numerical issues when computing stress differentials near repeated σ_i configurations due to arbitrarily small denominators of the form $\sigma_i \pm \sigma_j$ with potentially negated singular values. In addition, [SHST12] proves the existence of unavoidable energy ‘kinks’ at the same configurations, which cause force discontinuity. However, the dramatic distortion of the force field directions and its inversion-increasing effects, caused by the rapidly rotating singular vectors

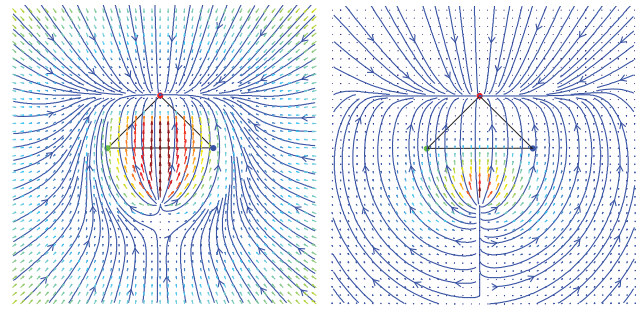


Figure 5: Non-linear force fields experienced by node x_1 (red) in a $[-3, 3]^2$ region of the plane when x_2 (green) and x_3 (blue) are fixed. Left and right panels show C^0 and C^1 extrapolations of the Neo–Hookean hyperelastic material from [SHST12], with the same parameters given there, $\varepsilon = 0.4$, $k = 20 \times E$. Material parameters: $E = 1000$, $\nu = 0.35$.

around such configurations is, to the best of our knowledge, unmentioned in invertible FEM research [ITF04], [TSIF05], [ST08], [GKJD09], [SZL*11], [SHST12], [SSB13].

As a conclusion, SVD schemes allow robust degeneration treatment for generic hyperelastic materials, difficult to achieve by other means, but have several drawbacks that discourage their use in inter-active simulations. Force continuity through collapse is guaranteed if only the smallest σ_N is zero, which is the most common case, but lost when additional principal stretches become 0 or the configuration passes through a critical point. Self-correction cannot be guaranteed and equilibrium inverted configurations exist. We show the macroscopic effects of these flaws in Section 6.

5. Computing a Robust, Time-Consistent and Self-Correcting $\mathcal{R}$

After analysing the existing degenerate element treatment methods we identify the two main sources of problems:

- Discrete-time heuristic degeneration direction: Any heuristic direction selection that only analyses an instantaneous element configuration is bound to fail for large deformations in a single simulation timestep (Figure 6). As we saw in Section 4.3, enforcing time-consistency of the degeneration direction is not enough, and may consistently enforce a wrong recovery direction if the heuristic fails. Moreover, incorrect recovery directions may induce degeneration on neighbour elements.
- Existence of critical points in the computation of $\mathcal{R}$: Critical points in the rotation fields generated by QR and SVD dramatically distort the inferred $\mathcal{R}$ around them, spanning all possible orientations. In the presence of critical points, CLFEM intuitively wraps LFEM forces around them (Figure 3). Such distortion yields locally discontinuous and inversion-increasing forces that threaten both robustness and self-correction and may result in equilibrium inverted configurations.

We propose a new method that solves both problems in sequence. First we will show how to compute a chronologically correct and

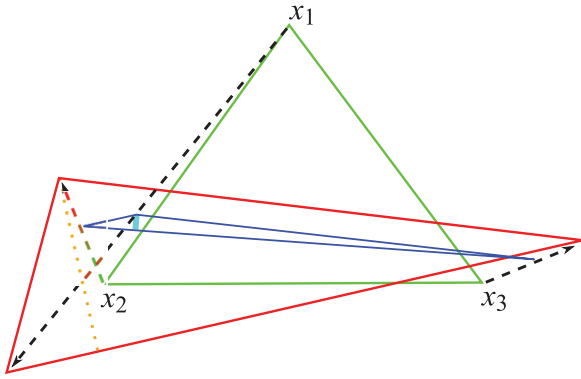


Figure 6: A 2D triangle element deforms from the green undegenerate configuration to the red inverted one along the dotted arrows in a single timestep. The configuration at the continuous time of collapse with $\det(\mathcal{F}) = \alpha$ is shown in blue, and implies that node x_1 collapsed to edge E_{23} (thick light blue line). An instantaneous heuristic would only analyse the final configuration and identify a shortest distance, in dotted orange, implying the collapse of node x_2 to edge E_{31} that would yield an incorrect uninversion direction.

time-consistent degeneration direction (Section 5.1) and afterwards we will explain how to use such direction to compute an $\mathcal{R}$ that avoids critical points and enforces robust and plausible recovery (Section 5.2).

5.1. Computing the chronologically correct degeneration direction

We observe that for a previously undegenerate simplex element in two dimensions (triangle) or three dimensions (tetrahedron) to become degenerate during a simulation timestep $[t_k, t_{k+1}]$, it must collapse with $\det(\mathcal{D}_s) = 0$ at some instant $t_c \in (t_k, t_{k+1}]$. We can analyse the element configuration at t_c to determine the degeneration direction $\hat{d}_c$, and use it to revert the degeneration in a chronologically correct and time-consistent way. This is not true for non-simplex elements (e.g. pyramids, prisms, bricks), which may degenerate without collapsing and would require a more complex treatment.

5.1.1. Computing t_c

Inspired by continuous collision detection techniques, in order to compute the *time of collapse* t_c we normalize the time interval to $\lambda \in [0, 1]$, assume linearly interpolated node trajectories $x_i(\lambda) = x_i(t_k) + \lambda(x_i(t_{k+1}) - x_i(t_k))$ and solve the polynomial equation $\det(\mathcal{D}_s(\lambda)) = 0$ to obtain a $\lambda_c \in (0, 1]$ equivalent to t_c . Node trajectories can always be observed and analysed regardless of their causes (e.g. dynamics, user interaction, scripted animation).

Equation $\det(\mathcal{D}_s(\lambda)) = 0$ is quadratic in two dimensions and cubic in three dimensions. Closed form solutions for the quadratic and cubic polynomial roots are available, however, we do not need to handle all possible cases, because λ_c is only computed when an actual crossing $\det(\mathcal{D}_s(1)) \leq 0 < \det(\mathcal{D}_s(0))$ is detected. We pur-

posefully ignore the possibility of an even number of roots (2) inside $(0, 1]$, as they represent self-corrected degenerations. Therefore, in two dimensions we are only interested in a single real root. In three dimensions, we may have either one or three real roots for $\lambda_c \in (0, 1]$ in case of crossing. If three roots exist, we are only interested in the largest $\lambda_c \leq 1$, as the previous two represent a self-corrected degeneration. We can similarly solve for $\det(\mathcal{D}_s(\lambda)) = K$ for any parameter K . This is mostly useful to find the λ_α at which the scale-independent element volume change fraction $\det(\mathcal{F}(\lambda)) = \det(\mathcal{D}_s(\lambda)) / \det(\mathcal{D}_m)$ falls below a certain threshold α by setting $K = \alpha \det(\mathcal{D}_m)$.

5.1.2. Computing $\hat{d}_c$

Once λ_c is known, the collapse configuration $x_i(t_c)$ can be explicitly computed and analysed to determine a chronologically correct and time-consistent *degeneration direction* $\hat{d}_c$ that will be used in successive timesteps until the element returns to an undegenerate configuration. This direction is not constant, but derived from the deforming element geometry. Time-consistency is achieved by storing a *collapse feature pair* (A_c, B_c) determined once at t_c , and computing the instantaneous $\hat{d}_c$ accordingly. As a convention, the sign of $\hat{d}_c$ will be chosen to point from A_c to B_c .

In three spatial dimensions a singular $\mathcal{F}$ is possible in the following basic collapse cases and their combinations:

- V-F: A vertex V_c collapses to its opposite face F_c .
- V-E: A vertex V_c collapses to an edge.
- V-V: Two vertices V_i and V_j collapse to a single point.
- E-E: Two edges E_{ij} and E_{kl} cross.

The highly improbable cases V-E and V-V can be included in the simple V-F case. In the V-E case, the edge must be part of F_c . In the V-V case we can arbitrarily choose one of the vertices as V_c and its opposite face as F_c . All three cases yield a collapse feature pair (V_c, F_c) and a degeneration direction $\hat{d}_c = \hat{N}_c$ normal to F_c . The E-E case is essential and results in a collapse feature pair (E_{ij}, E_{kl}) with degeneration direction $\hat{d}_c = \frac{e_{ij} \times e_{kl}}{\|e_{ij} \times e_{kl}\|}$.

In two dimensions only V-E and V-V cases are possible, and can be both reduced to a V-E pair (V_c, E_c) formed by a node and its opposite edge. To determine (V_c, E_c) we choose the E_c with largest squared length and use its normal as $\hat{d}_c$.

The determination of the collapse feature pair at the collapse configuration suffers from ambiguity in case of geometric near-coincidences, only amplified by finite floating-point precision. As a result, slightly different configurations at t_c may result in different feature pairs being chosen. However, as the element is in a flattened configuration, all candidate feature pairs (A_c, B_c) , regardless of their type, would yield equivalent $\hat{d}_c(t_c)$ perpendicular to the containing plane and following the sign convention to point from A_c to B_c . This includes the arbitrary transformation of V-V into V-F in three dimensions and V-V into V-E in two dimensions. As the element deforms the instantaneous $\hat{d}_c$ would diverge for the different candidate feature pairs, but this does not threaten the smoothness of the dynamics once a specific feature pair is selected among them. If such ambiguity was a serious concern, robust geometric predicates

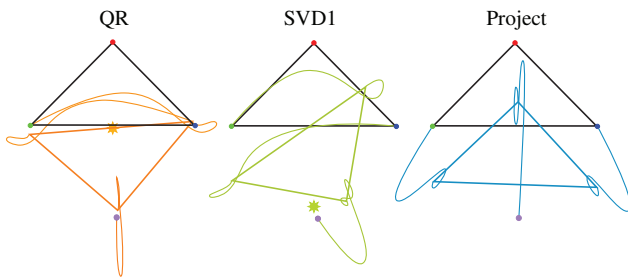


Figure 9: A triangle element is initially inverted by moving node x_1 (red) to the violet point behind edge E_{23} , which is initially kept in reference configuration, shown in black. Then, all three nodes of the element are freed and simulated until equilibrium, and their trajectories plotted. Left and middle panels show that the element resolves inversion and stabilizes by performing an unexpected large rotation due to the critical points in QR and SVD, represented as stars. The rightmost panel shows that Project avoids rotation, yielding much more plausible trajectories and stable configuration. Reflect produces a similar result. Material parameters: $E = 1000$, $\nu = 0.35$, damping ratio = 0.25.

Continuity: The continuity of $\tilde{\mathcal{R}}$ depends on the continuity of $\hat{d}_c$, which is guaranteed as long as none of the features used to derive it collapse. Such feature collapse requires the element to degenerate in a specific way while already inverted, a much less common *second degeneration*. In this case $\mathcal{R} = \mathcal{I}$ could be chosen, but we can improve continuity if we store the $\mathcal{R}$ from the previous timestep and reuse it until $\hat{d}_c$ is well defined again. This is still discontinuous, but will only affect the simulation shortly, typically for a single timestep. Similar discontinuous definitions of $\mathcal{R}$ are used in [ITF04] and [PO09] for highly degenerate configurations, and we observed no adverse effects in this specific case.

Differentiability: The rotation $\tilde{\mathcal{R}}$ is differentiable everywhere except (i) at the previously mentioned discontinuous configuration and (ii) across the transitions between undegenerate/degenerate configurations due to the switch between regular Polar Decomposition and Project/Reflect. This would be an issue in a fully implicit method, but for quasi-implicit integration, the approximation $\frac{\partial \mathcal{R}}{\partial x_e} \approx 0$ yields a constant force jacobian $\frac{\partial f_e}{\partial x_e} \approx -\mathcal{K}_e^{\mathcal{R}}$ that avoids non-differentiability problems.

Multiple inversion: For interactive simulation it would be impractical to require strict causality preservation. Therefore, when an inverted element uninverts in a chronologically incorrect way due to external factors, we consider it successfully recovered. This does not affect the continuity and differentiability of $\mathcal{R}$ across inversion/uninversion transitions, because once a witness feature pair (A_c, B_c) is selected at t_c , it is unconditionally used to compute $\hat{d}_c$ until the element becomes undegenerate, regardless of the actual uninversion node trajectories.

Initial degeneration: In case of initially degenerate elements with no history, Project and Reflect must temporally switch to a discrete-time heuristic fallback method. As rotation correction works on a per-element basis, we simply mark such elements in an initialization

Table 1: Several objects are randomly distorted and their recovery time shown in seconds.

	#Deg. elem.	QR	SVD1	SVD2	Project	Reflect
Box32	16	0.72	0.27	0.37	0.23	0.26
Box50	24	0.64	0.30	0.21	0.19	0.20
Box128	60	0.64	0.22	0.18	0.17	0.12
Whale31	16	0.25	0.21	0.12	0.09	0.09
Duck43	22	0.29	0.48	0.46	0.36	0.28
Snail87	45	0.23	0.11	0.18	0.12	0.09
Cube384	190	1.06	0.33	0.52	0.30	0.29
Cube1296	630	0.67	0.31	0.28	0.30	0.21
Bunny1085	537	0.60	0.37	0.32	0.26	0.22

phase and process their rotations with the SVD1 method until they recover.

As a result, the elastic forces generated by the Project and the Reflect methods strongly encourage elements to revert degeneration along the chronologically correct direction, instead of causing implausible local rotations due to critical points in QR and SVD. Figure 9 shows representative node trajectories for an isolated degenerate element. A comparison between QR, SVD1, SVD2, Project and Reflect methods will be presented in the next section.

6. Results

We implemented quasi-implicit CLFEM with corrected $\tilde{\mathcal{R}}$ computed using degenerate element treatment methods based on [NPF05] (QR), [ITF04] (SVD1) and [ST08] (SVD2), as well as our Project and Reflect methods.

Robustness is analysed in two stress tests, with results shown in Tables 1 and 2. These tests used a small set of representative objects: Box32, Box50 and Box128 are regular triangle discretizations of a square box with side length 1 and density $\rho = 10$ and Poisson ratio $\nu = 0.4$. Cube384 and Cube1296 are their 3D equivalents. Whale31, Duck43 and Snail87 (Figure 10) are spline-based 2D shapes with a longest extent of one discretized using Constrained Delaunay Triangulation to guarantee high element quality. The Stanford Bunny was tetrahedralized into 1085 elements using the TetGen software. All animal shapes have uniform material parameters $\rho = 10$, $\nu = 0.25$. Without loss of generality we use a soft material (Young modulus $E = 1000$) in all examples to easily demonstrate large deformations and degeneration. The number in each object's name specifies its element count. All objects were simulated using quasi-implicit Euler integration with a timestep of 1/60 s divided in four substeps in order to reduce the effects of numerical damping. The system of linear equations is solved using the iterative CG method with a target error residual of 10^{-3} and a maximum of 10 iterations.

The first stress test focuses on robust recovery from extreme situations. The results are shown in Table 1. The objects are initially distorted by moving their nodes to random positions inside their bounding box until roughly half of the elements become degenerate. For each object and method we show the time required to recover all

Table 2: Several objects in free fall impact the ground. We show the maximum number of degenerate elements and the time (in seconds) required to recover completely or stabilize, with unrecovered element count shown between parentheses.

	QR	SVD1	SVD2	Project	Reflect
Box32	12	7	7	7	7
	4.0	0.15	0.19	0.15	0.15
Box50	15	13	11	13	13
	1.11	0.55	0.54	0.55	0.55
Box128	45	29	24	29	29
	4.6	3.5 (3)	1.5 (2)	2.5 (3)	0.45
Whale31	11	8	9	8	8
	0.34	0.12	0.27	0.26	0.25
Duck43	16	13	14	14	14
	∞	0.67	0.66 (1)	0.65	0.66
Snail87	31	22	27	25	27
	0.67 (3)	0.95 (3)	0.29	0.25	0.29
Cube384	116	38	30	32	32
	1.8 (5)	0.6	0.58	0.59	0.58
Cube1296	363	168	156	160	190
	3.8 (27)	2.4	3.0 (9)	1.9	1.75
Bunny1085	287	160	141	126	125
	3.73 (54)	3.05 (35)	3.07 (39)	3.01 (21)	2.53 (19)

degenerate elements, in seconds. All methods successfully recover all objects pretty fast, with QR coming out as the slowest in all but one case and Project/Reflect consistently yielding the best results. For all methods we start the simulation with the same distorted configuration, but for SVD2, Project and Reflect the initial distortion trajectories were kinematically enforced using continuous time to allow the computation of meaningful degeneration directions.

The second stress test focuses on a violent impact situation, with results shown in Table 2. The objects are released into free fall from a height of 1 under a strong gravity acceleration ($g = -50$ for Box shapes, $g = -100$ for animal shapes) and simulated until they impact the ground and stabilize. The results show that QR fails in many cases, either by taking a long time to recover or never doing so (Duck43 does not stabilize with QR). SVD1 behaves better but results in long recovery times and incomplete recovery of Box128 and Snail87 cases. SVD2 compares quite well to SVD1, but yields visible jitter due to force discontinuity. Project improves over SVD1 but still fails to recover Box128. Reflect is the only method that manages to consistently recover all cases fast and completely, except for the Bunny1085 case, in which it also yields the best results. If degenerate elements are not recovered rapidly and oppose a strong elastic resistance to further degeneration, gravity traps them under the weight of the object and remain degenerate forever.

Next, the effects of degenerate element elastic force magnitude and direction are analysed for two different discretizations of a beam, in two different support conditions (Figures 11 and 12). The beam sizes are 2.1×0.6 and its material parameters $E = 1000$, $\nu = 0.25$, $\rho = 100$.

In the first test the beam is supported by its lowest central nodes (in green) and bends under a gravity until it reaches a stable config-

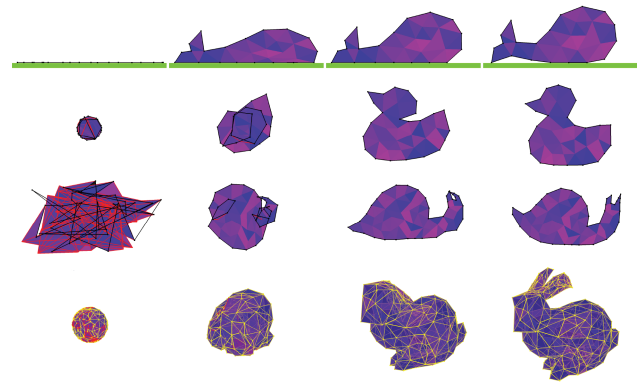


Figure 10: Several degeneration recovery tests using the Reflect method. Degenerate elements shown in red wireframe. Whale31 with $g = -10$, $\nu = 0.25$ and increasing Young modulus $E = 0$, $E = 50$, $E = 200$, $E = 1000$. Duck43 is trapped inside a ball and released at $t = 0$, shown at $t = 0.04$ s, $t = 0.3$ s and $t = 0.5$ s. Snail87 is randomly distorted and released at $t = 0$, shown at $t = 0.04$ s, $t = 0.14$ s and $t = 0.5$ s. Bunny1085 is trapped inside a ball at $t = 0$ and released, shown at $t = 0.01$ s, $t = 0.05$ s and $t = 0.1$ s.

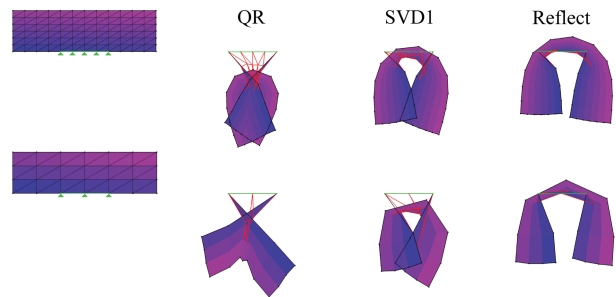


Figure 11: A centrally supported beam bends under gravity. Support nodes and edges shown in green, degenerate triangles are shown in red wireframe.

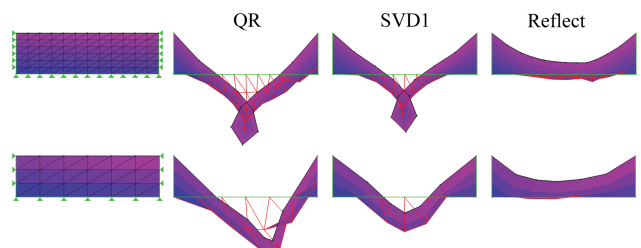


Figure 12: A supported beam collapses under gravity. Support nodes and edges are shown in green, degenerate triangles are shown in red wireframe.

uration (Figure 11). We increase gravity until all methods resulted in a visibly self-intersecting boundary. Reflect was the last method to collapse for both discretizations, at $g = -40$ for the coarse one, and near $g = -30$ for the finer one. Both QR and SVD1 collapse at lower g and yield visibly worse configurations than Reflect at the same g . Project behaves similar to Reflect. SVD2 resists collapse

Table 3: We simulated a single element during 10^6 explicit timesteps in two and three dimensions. For each dimension, in the first simulation the element never degenerates, in the second one the element spends roughly 50% of the timesteps inverted. For each method, we show the average CPU time spent on a single rotation extraction (in nanoseconds) and the percentage of the total computation time it represents.

	QR	SVD1	SVD2	Project	Reflect
Undeg.2D	14(4%)	55(14%)	55(14%)	31(8%)	31(8%)
Deg.2D	14(4%)	62(15%)	77(18%)	60(15%)	59(15%)
Undeg.3D	75(7%)	345(27%)	345(27%)	195(17%)	195(17%)
Deg.3D	75(7%)	350(27%)	378(29%)	225(20%)	215(19%)

up to almost the same g as Reflect, but nodes jitter around the stable configuration due to discontinuous forces.

In the second situation the beam is fixed at all boundary nodes except the top ones and we increase g up to -100 for all methods (Figure 12). The stronger and better oriented inversion resistance forces produced by Project/Reflect avoid structure collapse in situations were SVD1 and QR fail. As we increase gravity and some lower elements fail due to compression and degeneracy, QR and SVD1 forces push them downwards, in the wrong direction, and therefore increase the degeneration locally. Even worse, this effect propagates to neighbour elements and starts a failure ‘wave’ that collapses the whole object. In QR any minor inversion may trigger this process, as can be inferred from Figure 2. In SVD1 the element needs to invert more significantly, in the length-scale of its own height, to reach a configuration nearby the critical point, but once it does the propagation starts inevitably. Figure 1 shows analogous results in three dimensions.

6.1. Results analysis

The results show that the selection of a degenerate element treatment method has significant impact on the robustness, speed and smoothness of the correction process. Most importantly, the whole object dynamics and stable configuration in presence of degenerate elements are qualitatively different. Our methods improve both quantitative and qualitative results thanks to strong and consistently oriented degeneration reverting elastic forces. Among our methods, Reflect yields slightly better results than Project due to stronger un-inversion forces. However, Project does not require using SVD for collapsed elements, simplifying implementation.

The equilibrium inverted configurations near critical points found in QR (Section 4.1) and SVD (Section 4.3) approaches are essentially ‘impossible’ to achieve during numerical simulation with finite precision floating-point numbers. In most simulation scenarios, only a few elements will become degenerate at any time, and any potential flaws in their behaviour will be filtered by both damping and undegenerate neighbour elements. However, we have shown that the behaviour around critical points is sufficiently bad and probable to be considered an actual problem, and may result in macroscopic errors that are particularly serious for coarse meshes, common in interactive simulations.

We show some profiling results in Table 3. While the actual figures are strongly dependent on the specific implementation (custom C++), compiler (GCC 4.6.3 with $-O2$ optimization) and test hardware (Intel Core i5-2500K at 3.3 GHz), the relative performance of each method is still meaningful, and shows that both Project and Reflect are faster than SVD1 and SVD2 and slower than QR (which can handle degenerate elements without additional cost), but deliver the highest quality results among all methods.

7. Conclusions

We have shown that previously dismissed and unreported flaws in existing methods for handling individual degenerate elements may result in macroscopic failure of a whole object.

We propose a simple and efficient solution that avoids critical points and their associated flaws in existing approaches, extending the range of well-behaved configurations and yielding faster and smoother recovery from degeneration. This solution is specifically tailored for corotational linear elasticity and can be easily added to existing explicit/quasi-implicit CLFEM solvers with minor computational overhead. As a limitation, it cannot handle arbitrary hyperelastic materials as opposed to SVD-based methods, but avoids their drawbacks, making it superior for interactive simulations in which fast, robust and glitch-free behaviour is prioritized over material model accuracy or flexibility.

We plan to extend our method to anisotropic and plastic linear materials, and investigate the generalization to non-linear materials. We also consider applying our degeneration treatment to minimize the need for remeshing of bad-shaped elements in interactive cutting and fracture. Finally, we shall consider the extension to fully implicit integration.

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Supporting Information

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